

CEPHAN WATER SOLUBLE POWDER 500MG/G

Ingredient(s)
Each gram contains:
Cephalexin500mg
Sodium benzoate (as preservative).....1mg

Pharmacodynamics
Cephalexin is a first generation cephalosporin. Cephalexin acts by inhibiting the nucleopeptide synthesis of the bacterial wall. Cephalosporins interfere with transeptidation by acylating the enzyme making it unable to cross-link muramic acid-containing peptidoglycan strands. The inhibition of the biosynthesis of the material required to build the cell wall results in a defective cell wall and consequently osmotically unstable protoplasts. The combined action results in cell lysis and filament formation.
Cephalexin is active against Gram positive and Gram negative bacteria such as *Staphylococcus spp.* (including penicillin resistant strains), *Streptococcus spp.*, and *Escherichia coli*. Cephalexin is not inactivated by lactamases produced by Gram positive bacteria. However, beta-lactamases produced by gram-negative bacteria can inhibit cephalexin by hydrolysis of the beta-lactam cycle.
Resistance to cephalexin may be due to one of the following mechanisms of resistance. Firstly, the production of various beta-lactamases (cephalosporinase), that inactivate the antibiotic, is the most prevalent mechanism among gram negative bacteria. Secondly, a decreased affinity of the PBPs (penicillin-binding proteins) for beta-lactam drugs is frequently involved for beta-lactam resistant gram-positive bacteria. Lastly, efflux pumps, extruding the antibiotic from the bacterial cell, and structural changes in porins, reducing passive diffusion of the drug through the cell wall, may contribute to improve the resistant phenotype of a bacterium.
Well-known cross-resistance (involving the same resistance mechanism) exists between antibiotics belonging to the beta-lactam group due to structural similarities. It occurs with b-lactamases enzymes, structural changes in porins or variations in efflux pumps. Co-resistance (different resistance mechanisms involved) has been described in *E.coli* due to a plasmid harbouring various resistance genes.

Pharmacokinetics
Dogs
After single oral administration of the recommended dosage of 15 mg of cephalexin per kg of bodyweight to Beagle dogs, plasma concentrations were observed within 30 minutes. The plasma peak was observed at 1.3 hour with a plasma concentration of 18.2 mg/ml. The bioavailability of the active was over 90 %. Cephalexin was detected until 24 hours after the administration. The first urine specimen was collected within 2 to 12 hours with peak concentrations of cephalexin measured at 430 to 2758 mg/ml within 12 hours. After repeated oral administration of the same dosage, twice a day for 7 days, plasma peaks occurred 2 hours later with a concentration of 20 mg/ml. Over the treatment period, concentrations were maintained above 1 mg/ml. The mean elimination half-life is 2 hours. Skin levels were around 5.8 to 6.6 mg/g, 2 hours after treatment.
Cats
A single oral administration of 15 mg of cephalexin per kg of bodyweight in cats led to a bioavailability of 56 %. The plasma peak was observed at 1.55 hour following administration with a plasma concentration above 15.1 mg/ml. The mean plasma half-life was about 1 to 2 hours. The first urine specimen was collected between 4 and 24 hours with the highest concentrations ranging between 63.7 and 393 mg/ml, occurring within 24 hours. With the same dosage administered over 7 days, twice a day, the highest urine concentration of cephalexin reached between 518 and 1256 mg/ml.

Indication
For the treatment of bacterial skin infections(including deep and superficial pyoderma)s in dogs caused by organisms susceptible to cephalexin. For the treatment of cutaneous and subcutaneous infections (wounds and abscesses) in cats caused by organisms susceptible to cephalexin. For the treatment of urinary-tract infections (including nephritis and cystitis)in cats and dogs caused by organisms susceptible to cephalexin.

Target Species
Dogs and cats.

Recommended Dose
15 mg of cephalexin per kg of bodyweight twice daily (equivalent to 30 mg per kg of bodyweight per day) for duration of:
- 5 days in case of cutaneous and subcutaneous infections (wounds and abscesses) in cats;
- 14 days in case of urinary-tract infection in cats and dogs;
- At least 15 days in case of superficial infectious dermatitis in dogs;
- At least 28 days in case of deep infectious dermatitis in dogs.
To ensure correct dosage, bodyweight should be determined as accurately as possible to avoid underdosing. In severe or acute conditions, the dose may be doubled.

Mode of Administration
Orally via drinking water.
Contraindication
Do not use in animals which are known to be hypersensitive to penicillins and cephalosporins. Do not use in rabbits, guinea pigs, hamsters and gerbils.

Warning and Precautions
Special warnings for each target species
None Special precautions for use

Special precautions for use in animals
As with other antibiotics which are excreted mainly by the kidneys, unnecessary accumulation may occur in the body when renal function is impaired. In case of known renal insufficiency, the dose should be reduced and antimicrobials known to be nephrotoxic should not be administered concurrently. This product should not be used to treat puppies of less than 1 kg of bodyweight or kittens under 9 weeks of age. Use of the product deviating from the instructions given may increase the prevalence of bacteria resistant to cephalexin and may decrease the effectiveness of treatment with other cephalosporins and penicillins, due to the potential for cross-resistance. Local treatment of cutaneous and subcutaneous infections in cats should be considered as a complement of the antibiotic treatment. Use of the product should be based on susceptibility testing of the bacteria isolated from the animal. If this is not possible, therapy should be based on local epidemiological information.

Special precautions for the person administering the veterinary medicinal product to animals
Penicillins and cephalosporins may cause hypersensitivity (allergy) following injection, inhalation, ingestion or skin contact. Hypersensitivity to penicillins may lead to cross-reactions to cephalosporin and vice versa. Allergic reactions to these substances may occasionally be serious. Do not handle this product if you know you are sensitised or if you have been advised not to work with such preparations. Handle this product with great care to avoid exposure, taking all recommended precautions. Wash hands after use. If you develop symptoms following exposure such as skin rash, you should seek medical advice and show the doctor this warning. Swelling of the face, lips or eyes or difficulty in breathing are more-serious symptoms and require urgent medical attention.

Other precautions:
None
Interaction with Other Medicaments
None known

Pregnancy and Lactation
The product can be used in pregnant and lactating animals.

Side Effects
Transient episodes of soft faeces and vomiting may be observed in cats during treatment Hypersensitivity to cephalexin is rare; however, the product should not be administered to animals which are known to be hypersensitive to cephalexin or penicillins.

Environmental Property
None has been mentioned.

Symptoms and Treatment of Overdose
Trials performed on animals with up to 5 times the recommended dose of 15 mg/kg demonstrated that the product is well tolerated.

Storage Condition
Store at temperature below 30°C. Protect from direct sunlight and water.

Shelf Life
3 years. Used within 24 hours after reconstitution.

Withdrawal Periods
Not known.

Maximum Residual Limit (MRL)
Nil

Product Description
White to off-white powder
100g x100 packaging.

To declare Y.S.P. INDUSTRIES VIETNAM CO., LTD. as Manufacturer and Y.S.P. INDUSTRIES (M) SDN. BHD. as Product Registration Holder



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